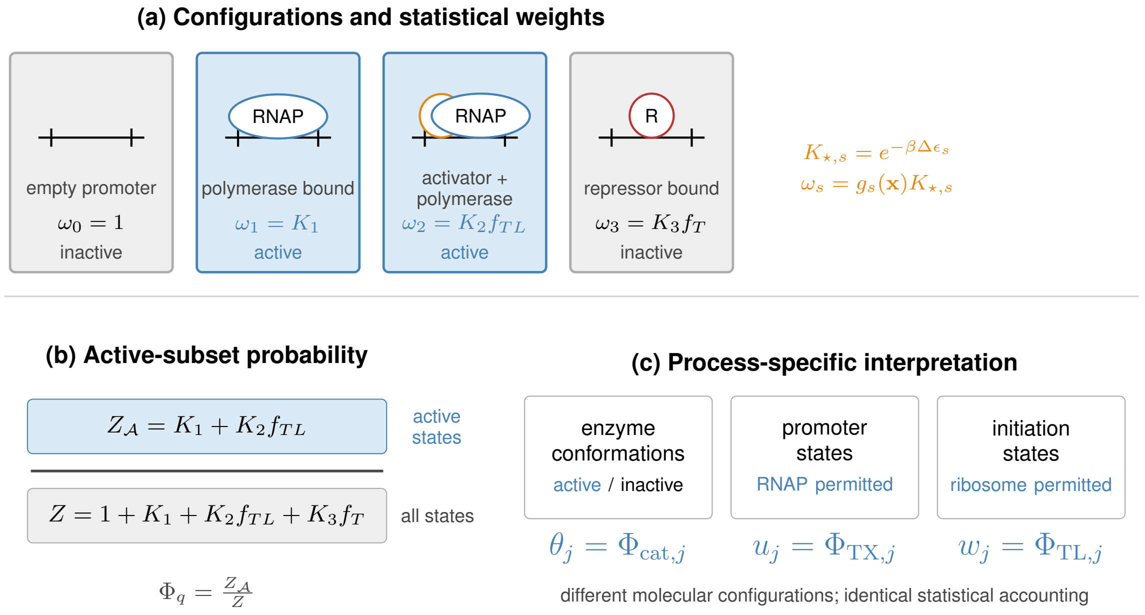
(a) Configurations and statistical weights
RNAP
RNAP
R
empty promoter
$\omega_0 = 1$
inactive
polymerase bound
$\omega_1 = K_1$
active
activator + polymerase
$\omega_2 = K_2 f_{TL}$
active
repressor bound
$\omega_3 = K_3 f_T$
inactive
$K_{\star,s} = e^{-\beta \Delta \epsilon_s}$
$\omega_s = g_s(\mathbf{x}) K_{\star,s}$
(b) Active-subset probability
$Z_{\mathcal{A}} = K_1 + K_2 f_{TL}$
active states
$Z = 1 + K_1 + K_2 f_{TL} + K_3 f_T$
all states
$\Phi_q = \frac{Z_{\mathcal{A}}}{Z}$
(c) Process-specific interpretation
enzyme conformations
active / inactive
promoter states
RNAP permitted
initiation states
ribosome permitted
$\theta_j = \Phi_{\mathrm{cat},j}$
$u_j = \Phi_{\mathrm{TX},j}$
$w_j = \Phi_{\mathrm{TL},j}$
different molecular configurations; identical statistical accounting